

Abdul Samad Shaik

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Profile Summary

Since 2019, I have been immersed in the world of Artificial Intelligence and Machine Learning, gaining hands-on experience across domains such as Image Processing, Computer Vision, Speech Processing, Time-Series Forecasting, LLM Fine-Tuning, and Model Optimization. I have trained numerous deep learning models from scratch using PyTorch and TensorFlow, continuously advancing my expertise in step with evolving AI trends. My work spans classical ML techniques and modern Agentic & Generative AI systems, focusing on developing, optimizing, and deploying models that deliver real-world impact.

EDUCATION

- **M.Tech.**, Communications, Signal Processing and Machine Learning | **IIT Dharwad** | CGPA: **8.9/10** 2023-2025
- **B.Tech.**, Electronics and Communication Engineering | **IIIT RGUKT RK Valley** | CGPA: **8.7/10** 2019 -2023

EXPERIENCE

- **Infobell IT Solutions Pvt Ltd** | AI Engineer | Jun 2025 – Till date
 - **Skills:** Agentic AI, RAG, MongoDB, ML Inference Optimization
 - Implemented RAG based systems.
 - Built Voice Agents using frameworks LiveKit, Pipecat
- **Siemens Technology and Services Pvt Ltd** | Technical Intern | Nov 2024 – May 2025
 - **Skills:** Pytorch, Software Development, Docker
 - Explored Federated Learning for Domain Adaptation
 - Explored 2 domains: Object Detection in Computer Vision and Time Series Foundation Models
 - Trained large-scale models from scratch in a Federated Learning setup using PyTorch
 - Improved model performance by 10–20% through experimentation with advanced Federated Learning strategies.
 - Developed framework for Federated learning using IIH SDK

SKILLS

- **Key Skills** | Machine Learning, Deep Learning, Generative AI, Computer Vision, Speech Processing, Agentic AI
- **Model Architectures** | CNNs, RNNs, Transformers, Vision Transformers, Diffusion Models, LLMs
- **Optimization & Deployment** | Model Optimization, Compression, Knowledge Distillation, ONNX Runtime
- **Frameworks** | PyTorch, TensorFlow, Scikit-learn, Numpy, Pandas, LangChain, LangGraph, FastAPI
- **Tools & Miscellaneous** | Python, Git & GitHub, Shell Scripting, Docker

PROJECTS

Master's Thesis: Deep Learning Enabled Segmentation, Classification and Probability Risk Assessment of Cervical Cancer Through Feature Extraction

- Implemented Various deep learning Models for segmentation and classification task
- Designed a **novel DL architecture (MRF-DCN)** achieving ~96% classification accuracy with fewer learnable parameters compared to state-of-the-art models.
- Developed a **Pipeline for segmentation, classification, and probabilistic risk assessment** to improve early detection and prediction of cervical cancer progression.

Bachelor Thesis: Smart Fatigue Relieve System

- Designed a new system to get relief from tiredness integrating **computer vision** to capture human facial status & **Arduino** for controlling Fan/AC, playing relaxing music, and deploying a small robot to provide water.
- Gathered facial expression data from university students of total **400** (200 from boys and 200 from girls) to train the **YOLO** model for fatigue detection.

PUBLICATIONS

- Abdul Samad Shaik, Revanth Palaparathi, Sahitya Chinta, and Rahul Jashvantbhai Pandya, “A Comparative Study of Deep Learning Architectures for Cervical Cell Segmentation in Pap Smear Images,” Accepted at IEEE International Conference on Information Technology (InCIT 2025), to be published in IEEE Xplore.
- Abdul Samad Shaik, Shashaank Mattur Aswatha, Rahul Jashvantbhai Pandya, “Deep Learning Enabled Segmentation, Classification and Risk Assessment of Cervical Cancer,” *arXiv preprint arXiv:2505.15505*, 2025.

ACHIEVEMENTS

- Gate EC 2022 – 2757/54292
- Gate EC 2023 – 3552/45833
- Secured 1st Prize in Technical Fest conducted - IIIT RGUKT RK Valley
- JEE Mains 2019 | Percentile 98.7451 | 14504/1147125